

MAX17320

2S-4S ModelGauge m5 EZ Fuel Gauge with Protector, Internal Self-Discharge Detection, and SHA-256 Authentication

General Description

The MAX17320 is a 38 μ A I_Q stand-alone pack-side fuel gauge IC with protector, battery internal self-discharge detection, and optional SHA-256 authentication for 2 to 4 series lithium-ion/polymer batteries.

The IC monitors the voltage, current, temperature and state of the battery to provide protection against over/undervoltage, overcurrent, short-circuit, over/undertemperature, overcharge, and internal self-discharge conditions using external high-side N-FETs, and provides charging prescription to ensure that the battery operates under safe conditions, thereby prolonging the life of the battery. The IC balances the cells using internal FETs. During prequal charging, the IC uses the CHGFET to linearly control charging.

To prevent battery pack cloning, the IC integrates SHA-256 authentication with a 160-bit secret key. Each IC incorporates a unique 64-bit ID.

The fuel gauge uses Maxim ModelGauge™ m5 algorithm that combines the short-term accuracy and linearity of a coulomb counter with the long-term stability of a voltage-based fuel gauge to provide industry-leading fuel gauge accuracy. The IC automatically compensates for cell aging, temperature, and discharge rate, and provides accurate state-of-charge (SOC) in milliampere-hours (mAh) or percentage (%) over a wide range of operating conditions.

The IC provides a configurable always-on LDO that can power small critical loads on the system side without overloading the cells even under fault conditions. Dynamic power functionality provides the instantaneous maximum battery output power which can be delivered to the system without violating the minimum system input voltage.

A Maxim 1-Wire® or 2-wire I²C/SMBus interface provides access to data and control registers. The IC is available in a lead-free, 2.4mm x 2.6mm 30-bump 0.4mm pitch WLP and 4mm x 4mm 24-pin TQFN packages.

Applications

- Smartphones, Tablets, and 2-in-1 Laptops
- Medical Devices, Health and Fitness Monitors
- Digital Still, Video, and Action Cameras
- Handheld Computers, Radios, and Terminals
- Power Tools, Wireless Speakers, and Drones
- Smart Batteries and Battery Backup

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Benefits and Features

- Battery Health + Programmable Safety/Protection
 - Overvoltage (Temperature Dependent)
 - Overcharge/Overdischarge/Short-Circuit Current
 - Over/Undertemperature
 - Undervoltage + SmartEmpty
 - Battery Internal Self-Discharge Detection
 - Ideal Diode Discharge During Charge Fault
 - Charging Prescriptions (JEITA)
 - Prequal Charge Control with CHG FET
- ModelGauge m5 EZ Algorithm
 - Percent, Capacity, Time-to-Empty/Full, Age
 - Cycle+™ Age Forecast
- Precision Measurement Without Calibration
 - Current, Voltage, Power, Time, Cycles
 - Die Temperature, 4 Thermistors $\pm 1^\circ\text{C}$
- Cell Balancing with Internal FETs
- SHA-256 Authentication
- Nonvolatile Memory for Stand-Alone Operation
 - History Logging, User Data (84 Bytes)
- Low Quiescent Current
 - FETs Enabled: 38 μ A Active
- Pushbutton Wakeup—Eliminates System Consumption Until Button Press
- Always-On LDO
- Dynamic Power—Estimates Power Capability
- SBS 1.1 Compatible Register Set

Simplified Block Diagram

